Sampling and Reconstruction

ECSE 313 Group 16 Lab 4 Report

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1 Sampling and Reconstruction Using Sample-and-Hold

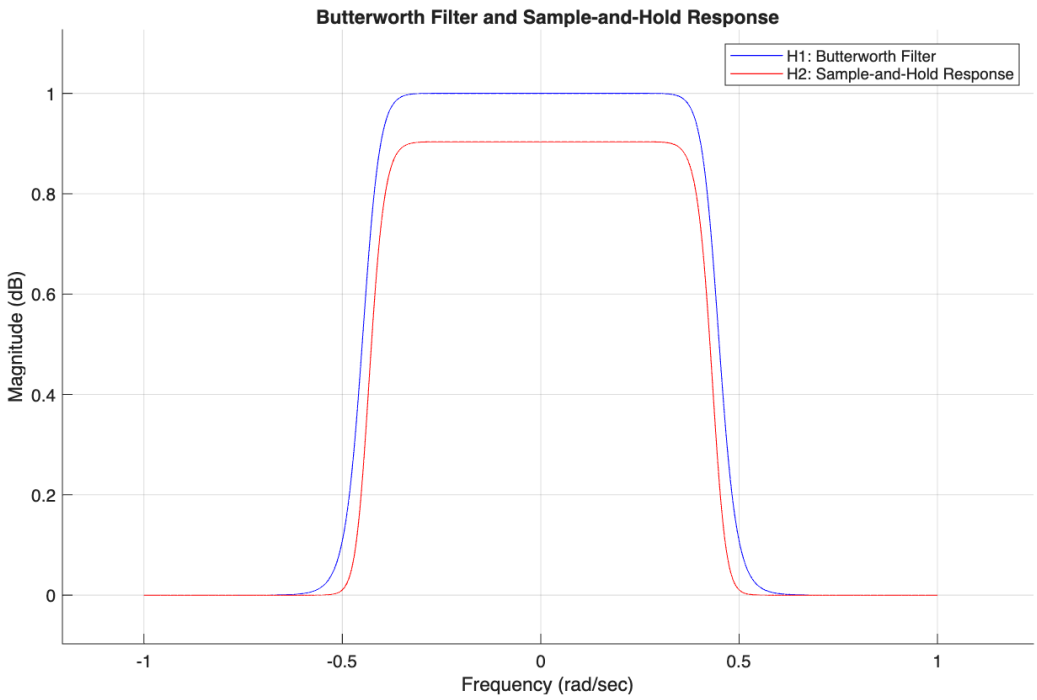

Figure 1: Magnitude Response of Butterworth Filter and Sample-and-Hold Response

The magnitude of the Butterworth filter is greater than that with the sample-and-hold device. The magnitude response of the sample-and-hold acts as a low pass filter. For high quality audio CD players, a low pass filter smooths the signal and improves audio quality.

2 Sampling and Reconstruction Using An Impulse Generator

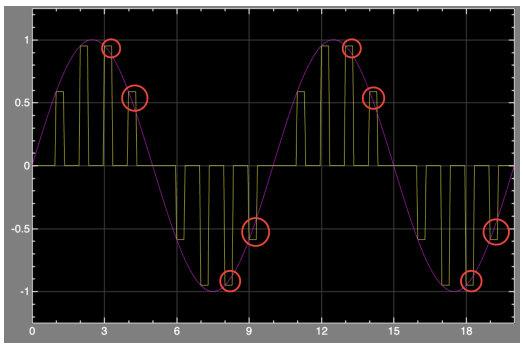

Figure 2: Input/Output Signals at 0.3 Sec Pulse

The circles in Figure 2 correspond to the aliasing of the reconstructed signal.

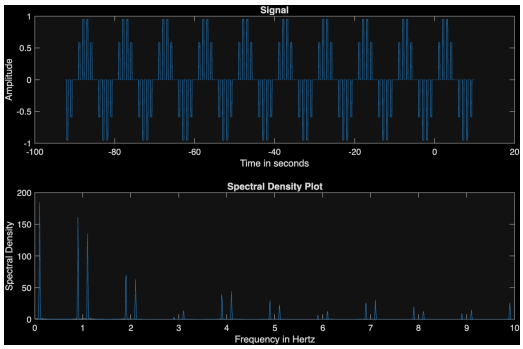

Figure 3: Output Signal and Frequency Spectrum at 0.3 Sec Pulse

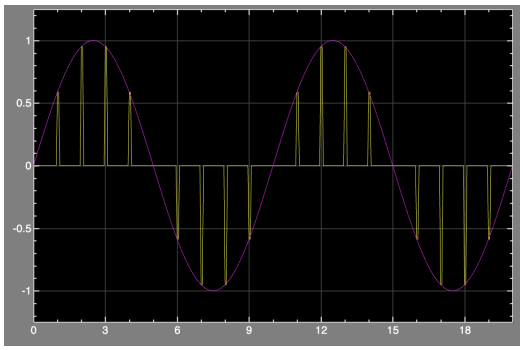

Figure 4: Input/Output Signals at 0.1 Sec Pulse

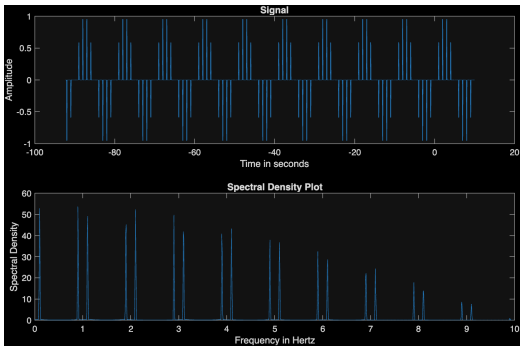

Figure 5: Output Signal and Frequency Spectrum at 0.1 Sec Pulse

The aliasing curves in Figure 4 have a smaller width than those in Figure 2. Additionally, the spectral density in Figure 5 has a lower magnitude intensity than that in Figure 3. These changes are caused by reducing the pulse width. When the pulse width is decreased, the average time the signal is "on" decreases and the spectral density decreases.

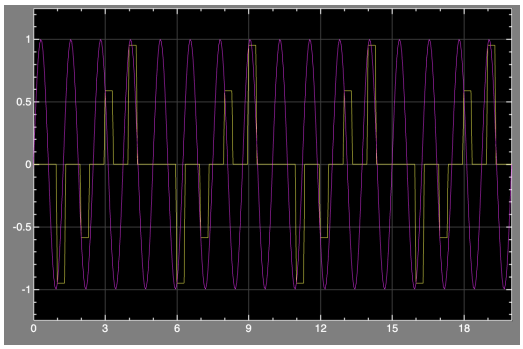

Figure 6: Input/Output Signals at 0.3 Sec Pulse with 0.8 Hz

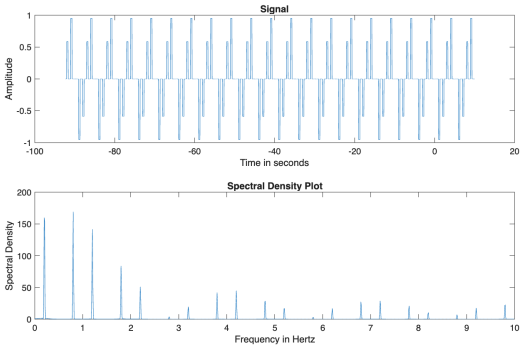

Figure 7: Output Signal and Frequency Spectrum at 0.3 Sec Pulse with 0.8 Hz

The frequency peak that corresponds to the lowest frequency (the fundamental component) of the output signal in Figure 7 is 2 Hz. The lowest frequency is not the same as the frequency of the input sinusoid because the lowest frequency is related to the Nyquist Frequency. The Nyquist Frequency is determined by the pulse width of 0.3 sec and is found by:

$$f = \frac{1}{2 \times T_s} = \frac{1}{2 \times 0.3sec} = 1.67Hz \tag{1}$$

Since aliasing occurs at frequencies greater than the Nyquist Frequency, the aliasing in Figure 7 begins at 2 Hz.

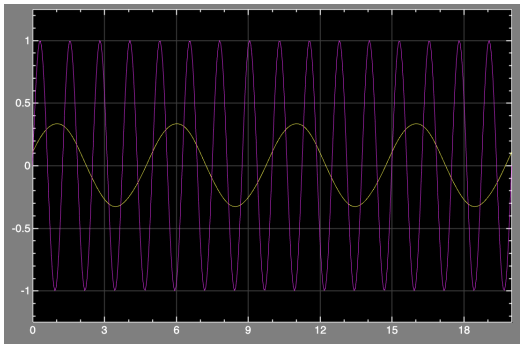

Figure 8: Input/Output Signals with Butterworth Filter

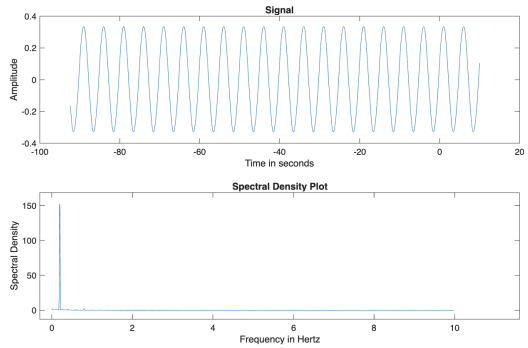

Figure 9: Output Signal and Frequency Spectrum with Butterworth Filter

The frequency of the Butterworth filter is determined by the cutoff frequency, which is 0.5 Hz. For frequencies greater than the cutoff frequency in Figure 9, the Butterworth filter acts as a stopband. Therefore, the frequency is 0 for frequencies greater than 0.5 Hz.

3 Sampling and Reconstruction with Sample and Hold

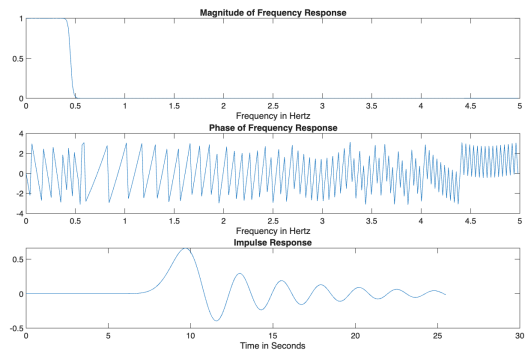

Figure 10: Sample-and-Hold With Two Scopes

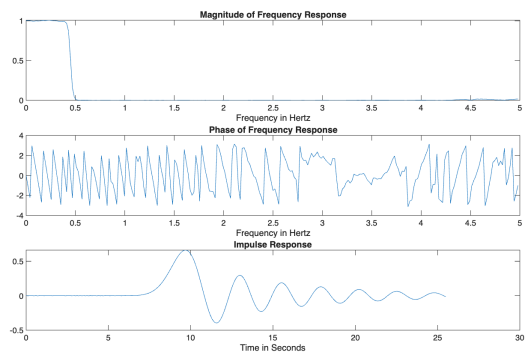

Figure 11: Sample-and-Hold With Two Scopes With Sample Time 1 Sec

The magnitude between Figure 10 and Figure 11 is nearly identical. However, for frequencies greater than 4.5 Hz and frequencies less than 0.45 Hz, Figure 10 has a magnitude of 0 while the magnitude in Figure 11 fluctuates. The sample and hold filter added in Figure 11 holds the signal constant until the next sample time. The fluctuations in Figure 11 represent the sample period of the filter. The analytical representation of this phenomena is described as:

$$p(t) = rect(\frac{t}{T_s} - \frac{1}{2}) = rect(\frac{0.45}{1} - \frac{1}{2}) \tag{2}$$

4 Discrete-Time Interpolation

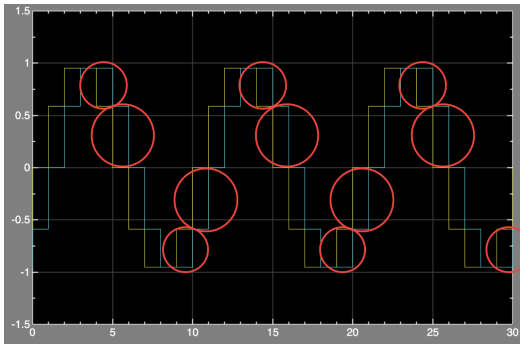

Figure 12: Discrete Time Interpolator Scope

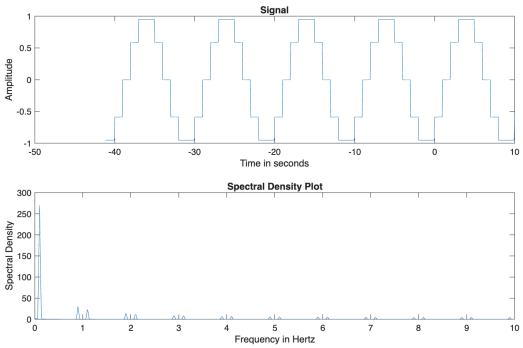

Figure 13: Signal.c and Frequency Spectrum

The circles in Figure 12 represent the aliasing of the signal.

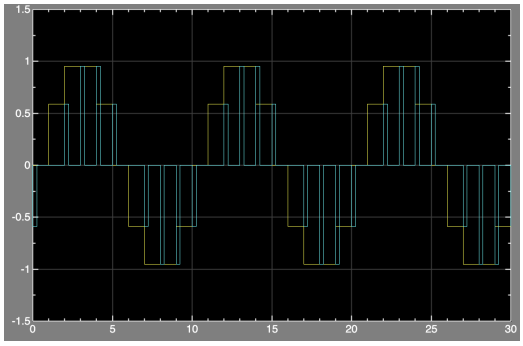

Figure 14: Discrete Time Interpolator Scope With 4 Hz

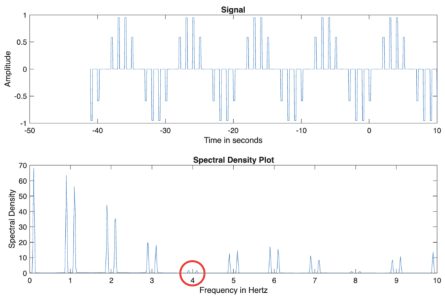

Figure 15: Signal.c and Frequency Spectrum With 4 Hz

The first aliased component and center frequency in Figure 14 is at 4 Hz because the spectral density suddenly decreases and the two peaks are centered around 4 Hz. The envelope shape for the spectral density represents sinc function of the Discrete-Time Filter.

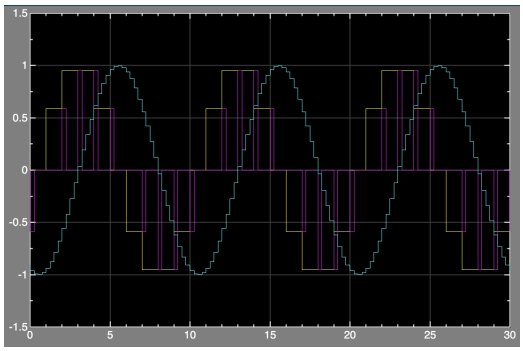

Figure 16: Discrete Time Interpolator Scope With 4 Hz

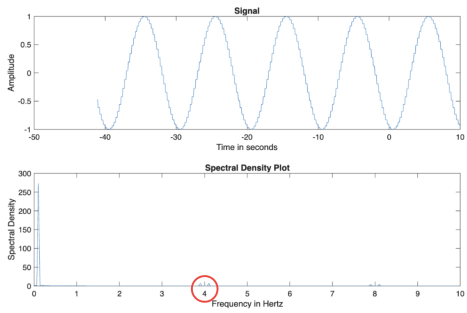

Figure 17: Signal.c and Frequency Spectrum With 4 Hz

The gain and cutoff frequency used for Figures 16 and 17 are 4 and $\frac{\pi}{4}$ respectively. These values clearly produce an output signal (blue curve) with the desired interpolation in Figure 16. The discrete-time interpolation is desirable before reconstructing a sampled signal because the sampling rate increases and spectral density of unwanted images in low-frequency ranges are reduced. Therefore, a more accurate representation of the signal is produced.

5 Discrete-Time Decimation

The quality of sig1 is better than the original audio at a sampling rate of 8kHz. However, both the sig1 audio produces choppy sound quality compared to sig2. There was a noticable distortion in the sig1 audio likely produced from interpolation. The sig2 audio, which uses the two decimation method, plays the cleanest sound of the three signals. Furthermore, the sample-and-hold process from the sig1 is reduced in the sig2 using a discrete-time interpolation.

audio.m

```
clc; clear; close all;

% Read in the music.au file
[y, fs] = audioread('music.au');

% Play the original audio
sound(y, fs);

% Downsample the audio
sig1 = y(1:2:end);

```

```
% Play the downsampled audio
sound(sig1, fs);

M = 20;
W = pi/2;
h = fir1(M, W/pi);

% Play signal with low-pass and discrete-time filter
filtered = conv(y, h);
sig2 = filtered(1:2:end);
sound(sig2, fs);
```